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Observable-Class Predictability in Stylised Three-Dimensional Self-Gravitating N-body Families: Coarse Dominance, Kinematic Advantage, and Softening Dependence

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ABSTRACT

We perform a controlled battery study of predictive performance across coarse positional, fine positional, and fine kinematic observables in four stylised three-dimensional self-gravitating N-body families, using 140000 simulations with 500 realisations per cell, two force models, and a five-point softening sweep ($\varepsilon \in [0.02, 0.10]$). The primary result is robust coarse dominance for merging binary clusters: grid-scale density variance predicts future clustering at $r = 0.984$ ($\varepsilon = 0.05$; winner-gap CI $[-0.42, -0.30]$) across all tested N, ε, and force models. A secondary result is a restricted fine-kinematic regime: local velocity dispersion outperforms the coarse predictor for concentrated cusp profiles at low softening (Hernquist $\varepsilon \leq 0.05$, Plummer $\varepsilon = 0.02$; gap CIs exclude zero), though the effect is modest ($R^2 \approx 0.18$–0.28 at $N = 1024$). This advantage erodes with increasing ε and vanishes by $\varepsilon = 0.10$. A held-out commonality re-analysis further shows that this kinematic advantage is predominantly a *bulk velocity-scale* effect: a single global velocity-scale scalar recovers 81–90% of the local-velocity-dispersion advantage, leaving only a small (∼9–19%) genuinely local residual—so the advantage is better read as sensitivity to the overall virial temperature of a realisation than as access to local phase-space structure. The strongest negative result is that positional fine-scale observables rarely outperform the coarse predictor: the positional-class gap CI excludes zero in only 10 of 80 direct-isolated cells (Hernquist at low ε and cold-clumpy at $\varepsilon = 0.10$), compared to the kinematic class which dominates the fine advantage wherever it appears. No universal ranking is supported: predictive advantage is structured by initial-condition family, observable class, and the softening-to-scale-radius ratio ε/a. The point is not that finer descriptions are generally worse, but that additional spatial detail alone is insufficient unless it carries the dynamically relevant information channel.

Keywords: N-body simulations — gravitational softening — phase-space structure — initial conditions

1. INTRODUCTION

This paper asks, in a controlled setting with stylised initial-condition families, which observable class—coarse positional, fine positional, or fine kinematic—best predicts future clustering. The systems are intentionally simplified (isolated or periodic, idealised IC families, fixed scalar target, no cosmological expansion), so the result is a statement about predictive structure in this controlled setting, not a general claim about self-gravitating systems. The goal is comparative: to map which observable class wins, and under what conditions, across a large battery of three-dimensional N-body simulations.

The Jeans instability criterion provides a physical prior. The Jeans length $\lambda_J \propto \sigma_v \rho^{-1/2}$ sets the scale above which gravitational collapse overcomes pressure support (Binney & Tremaine 2008). For a bimodal merger, λ_J is comparable to the inter-cluster separation ($\sim 0.5\,L$), far exceeding the grid scale, so the coarse density field should capture the decisive instability. For concentrated cusps (Hernquist, Plummer), λ_J near the core is small ($\sim a$); when the force resolution resolves this scale, kinematic substructure below λ_J may carry additional predictive information. For cold-clumpy configurations, the low internal dispersion yields a small λ_J within each clump, but the inter-clump scale again favours coarse predictors. These qualitative expectations motivate a systematic empirical test.

The predictive-summary approach taken here connects to a broader literature on collisionless relaxation in self-gravitating systems. After gravitational collapse, N-body systems reach quasi-equilibrium via *violent relaxation* (Lynden-Bell 1967): rapid fluctuations in the mean gravitational potential mix orbits in phase space and erase microscopic memory on a dynamical timescale, without conserving individual particle energies. Modern work has shown that this erasure is incomplete—coarse-grained phase-space distributions retain information about the initial conditions beyond what collisional two-body relaxation alone would predict (Chavanis 2002). Our framework tests the practical consequence: which scale of initial summary—large-scale density contrast or small-scale phase-space substructure—carries enough residual memory to predict future clustering. For a comprehensive review of N-body force algorithms and the resolution tradeoffs that frame this question, see Dehnen & Read (2011).

The question of which initial summaries survive into later structure has an operational dimension: it determines which statistics are worth computing at the initial-condition stage and which are dynamically irrelevant. Structure-formation studies show that merger remnants retain kinematic memory of their progenitors—shells, streams, and velocity anisotropy persist well after the density profile has relaxed (White 1996; Hernquist & Spergel 1992)—but whether that memory is accessible from the scalar summary statistics routinely computed in simulation pipelines is an open question. Our bimodal and concentrated-cusp families test, in a controlled and deliberately stylised setting, the same dichotomy: does the large-scale density contrast or the small-scale kinematic substructure carry more predictive power for future clustering?

We study four initial-condition families spanning these structural types (Section 3) with 140000 three-dimensional N-body simulations. For each we test observables belonging to three classes—coarse positional, fine positional, and fine kinematic—using Pearson correlation with bootstrap confidence intervals as the verdict criterion. The winner gap is defined as $\max|r_{\mathrm{fine}}| - \max|r_{\mathrm{coarse}}|$ throughout: positive values indicate fine advantage, negative values indicate coarse dominance. The key question is whether additional spatial detail beyond the coarse density field carries predictive power, or whether only the kinematic (phase-space) channel does so, and under what conditions.

2. MODELS AND NUMERICAL METHODS

2.1. Force models

A direct-sum isolated solver and a particle-mesh periodic solver are run across the same IC families; PM results serve as a numerical cross-check with a different force architecture and boundary conditions (Section 5.9).

Direct-sum isolated (`direct_isolated`). Forces are computed via the softened pairwise potential $\phi_{ij} = -Gm(r_{ij}^2 + \varepsilon^2)^{-1/2}$ at $O(N^2)$ cost. The domain is open. Grid-based observables (coarse density variance, small-scale Fourier power) restrict to particles inside $[0, L)^3$; distance-based observables (kNN, ClosePairs, VelDisp, FoF) use all particles via KDTree queries without box restriction.

Particle-mesh periodic (`pm_periodic`). The density field is deposited via cloud-in-cell (CIC) interpolation onto a 32^3 grid; the Poisson equation is solved spectrally by fast Fourier transform (Hockney & Eastwood 1988). Periodic boundary conditions are enforced. The PM Hamiltonian differs from the direct-sum Hamiltonian; PM results are reported as a cross-check in Section 5.9 rather than as a co-equal second measurement of the same physics.

2.2. Integrator

Both models use leapfrog KDK (kick-drift-kick) with fixed time step $\delta t = 0.005$, $G = 1$, unit total mass, and box size $L = 2$ (Quinn et al. 1997; Yoshida 1990). Total integration: 600 steps ($t_{\mathrm{final}} = 3.0$ in natural units). Snapshots at steps 0, 100, 300, and 600. The early horizon at step 100 ($t = 0.5$ in natural units, approximately $0.3\,t_{\mathrm{cross}}$ for the bimodal configuration) is chosen to capture the initial collapse phase before secular relaxation dominates: for the bimodal case this coincides with the first pericentric passage; for concentrated profiles it spans the initial violent relaxation transient (Lynden-Bell 1967).

2.3. Battery design

Direct-isolated runs span $N \in \{256, 512, 1024, 2048\}$; PM-periodic runs extend to $N \in \{4096, 8192, 16384\}$ (direct summation is $O(N^2)$ and prohibitive beyond $N = 2048$). Both models share the same ε grid $\{0.02, 0.03, 0.05, 0.07, 0.10\}$, four core IC families, and 500 independent realisations per cell. The softening parameter ε is treated as a force-resolution control variable: smaller values sharpen the interaction potential at the cost of larger integration errors for close approaches, while larger values suppress cusp dynamics by smoothing the potential over a physical scale (Dehnen 2001; Dehnen & Read 2011; Athanassoula et al. 2000). The five-point ε grid spans $\varepsilon/a \in [0.10, 0.50]$ for the concentrated families, bracketing the force-convergence regime identified by Power et al. (2003) as necessary for reliable cusp profiles. Each replicate uses a distinct random seed for particle placement drawn from the IC distribution; all other parameters are held fixed. Replicates therefore capture sampling variance in the initial structural observables, which drives variance in future clustering outcomes and makes the Pearson correlations estimable. In addition to the four core families, three angular-shuffle null variants (bimodal, Hernquist, and Plummer with randomised angular positions) serve as controls, yielding

7 families in total. This yields 280 parameter cells and 140000 total runs. Bootstrap CIs use 1000 resamples per cell.

The bimodal IC gives each sub-cluster an infall velocity of 0.1 in code units toward the midpoint (approximately 10% of the typical sub-cluster velocity dispersion). Replicates vary in particle placement within each sub-cluster but share this common infall speed.

The ClosePairs observable is defined as pairs within 4ε. Across the ε sweep this measures five different physical scales (0.08, 0.12, 0.20, 0.28, 0.40 in box units), which complicates direct cross-ε comparison for this observable relative to the others.

3. INITIAL CONDITION FAMILIES

Figure 1 shows all four families as projected density images for $N = 512$. The families span large-scale topology and inner-profile shape.

Bimodal (`bimodal3d`). Two equal-mass Plummer sub-clusters of scale radius $a = 0.10$ placed at $\pm 0.25\,L$ along the x-axis with an infall velocity of 0.1 code units toward the midpoint (approximately 10% of the sub-cluster velocity dispersion). Future collapse rate should be set by the inter-cluster density contrast, making coarse dominance the expected outcome from Jeans-scale arguments. This configuration serves as a calibration anchor: confirming the expected coarse win validates the battery methodology before examining the more scientifically interesting concentrated-profile families, where the outcome is not predicted a priori. This configuration models pre-merger galaxy pairs or interacting dark matter haloes.

Hernquist (`hernquist3d`). Drawn from $\rho \propto r^{-1}(r + a)^{-3}$ (Hernquist 1990) with $a = 0.20$, velocities from an isotropic Gaussian at $\sigma \sim (GM/6a)^{1/2}$ (order-of-magnitude virial estimate). The steep inner cusp concentrates kinematic energy at small radii. This profile is widely used to model elliptical galaxies and the stellar component of dark matter haloes.

Plummer (`plummer3d`). Drawn from $\rho \propto (r^2 + a^2)^{-5/2}$ (Plummer 1911) with $a = 0.20$. Softer core than Hernquist; less concentrated velocity field. Plummer spheres are standard models for globular clusters (Heggie & Hut 2003; Spitzer 1987) and the cores of dwarf spheroidal galaxies.

Cold-clumpy (`cold_clumpy3d`). Five equal-mass Plummer sub-clusters with random positions and cold internal dispersions ($\sigma_{\rm inner} = 0.05$). This configuration models young stellar groups, fragmenting gas clouds, or the early assembly of dark matter substructure.

4. OBSERVABLES AND STATISTICAL DESIGN

4.1. Observable classes

Coarse positional. Density variance on a regular g^3 grid, $\sigma_\rho^2(g) = \mathrm{Var}[\rho_{\mathbf{i}}]$, for $g \in \{4, 8, 16\}$. Primary coarse observable: $\sigma_\rho^2(8)$ (CoarseG8).

Fine positional (three observables). (i) kNN-all: mean k-nearest-neighbour density over all in-box particles at $k = 16$. (ii) Pk: power in Fourier modes with $|\mathbf{k}| > g/4$. (iii) ClosePairs: fraction of pairs within 4ε.

Fine kinematic. Local velocity dispersion $\sigma_v^{\rm loc}(i) = \langle \|\mathbf{v}_j - \bar{\mathbf{v}}_{\mathcal{N}(i)}\|^2 \rangle^{1/2}_{j \in \mathcal{N}_k(i)}$ averaged over all in-box particles (VelDisp). This is the only observable accessing velocity (phase-space) information; across realisations, however, it tracks predominantly the system's global velocity scale rather than local phase-space structure (Section 5.6).

Fine structural. FoF group count at linking length $b = 0.20\,\bar{d}$ (recorded; included in the summary matrix).

4.2. Prediction target and verdict criterion

Primary target: $\Delta C_8^{\rm early}$, the change in $\sigma_\rho^2(8)$ between $t = 0$ and step 100. Verdicts are based on the 95% bootstrap CI of the winner gap $g = |r_{\rm best\,fine}| - |r_{\rm best\,coarse}|$ (1000 resamples of the 500 replicates, percentile bootstrap intervals with null-bias correction): FINE if the CI lower bound $> +0.05$; COARSE if the CI upper bound < -0.05; TIE otherwise (threshold set a priori). Point-estimate winner gaps are reported for effect-size description but do not determine verdicts; a positive point estimate with a CI spanning zero yields a TIE. Pearson correlations are computed across the 500 replicates per cell.

Null-bias correction. Winner-gap CIs are corrected for null bias by permuting observable-class labels on the within-cell correlation matrix and subtracting the mean permutation-null gap from each bootstrap draw. This bias arises from unequal class sizes (more positional-fine predictors than coarse or kinematic predictors), which produces a positive expected gap under the null even when no class dominates. The correction shifts CI bounds by 0.05–0.15 in cells with strongly unequal class sizes; its magnitude is largest in cells with many positional fine predictors and smallest in bimodal cells where the coarse signal dominates unconditionally. Because the permutation is performed on the actual data covariance (not an independent null draw), strong predictors contribute to the null pool, which may modestly inflate the corrected kinematic fine advantage in concentrated profiles. The main qualitative verdicts do not depend on this correction alone: they persist when the coarse final state replaces the difference target (Section 4.2) and when verdicts are read directly from class-level gap distributions without correction.

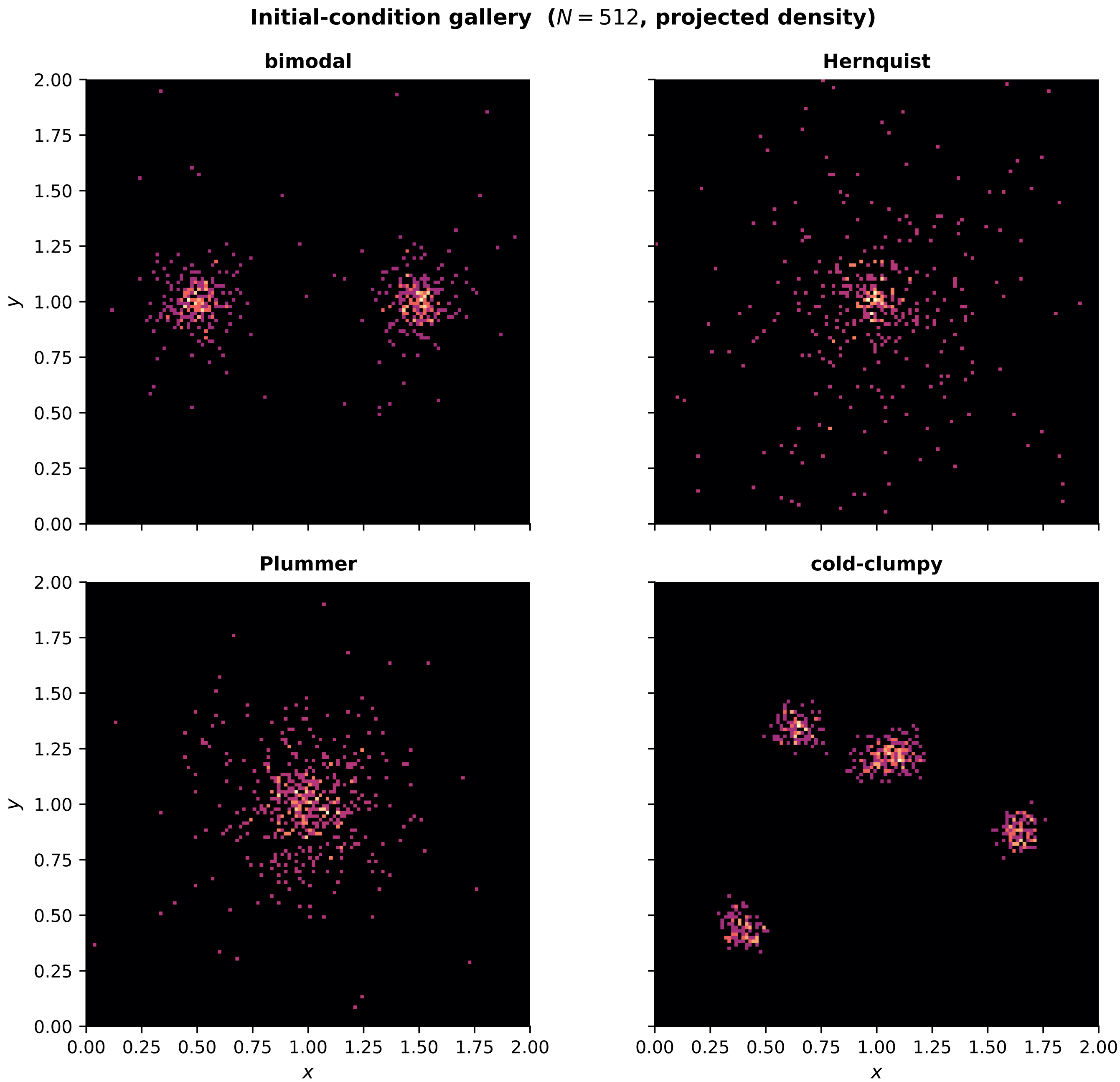

Figure 1. Initial condition families at $N = 512$, direct-isolated, $\varepsilon = 0.05$. Each panel shows projected density on a 128^2 grid (log scale). Left to right, top to bottom: bimodal (two-cluster), Hernquist, Plummer, cold-clumpy.

Shared-baseline note. The primary target $\Delta C_8^{\text{early}} = \sigma_\rho^2(8)|_{t=t_e} - \sigma_\rho^2(8)|_{t=0}$ shares the initial coarse density $\sigma_\rho^2(8)|_{t=0}$ with the coarse predictor. By the identity $\mathrm{Cov}(X, Y - X) = \mathrm{Cov}(X, Y) - \mathrm{Var}(X)$, this systematically deflates the coarse predictor's correlation with the difference target relative to its correlation with the final state alone. Fine predictors are not subject to this deflation. As a robustness check, repeating the analysis with $\sigma_\rho^2(8)|_{t=t_e}$ (the coarse final state, no baseline subtraction) as the target preserves the direction and significance of all reported verdicts, confirming that the bimodal coarse dominance and the concentrated-profile fine advantage are not artefacts of the difference formulation.

5. RESULTS

Throughout this section, r denotes the Pearson correlation of an initial observable with $\Delta C_8^{\text{early}}$. Comparisons use $|r|$ (predictive strength); the sign of r reflects the direction of the effect (e.g. $r < 0$ for bimodal coarse means higher initial density variance predicts more collapse, i.e. smaller $\Delta C_8^{\text{early}}$). The winner gap is defined as $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$; negative values indicate coarse dominance.

5.1. Numerical credibility

Table 1 summarises conservation diagnostics for all direct-isolated runs. The median relative energy drift is $4.48e-04$ (maximum $1.69e+01$). The maximum drift occurs in a single Hernquist run at $N = 2048$, $\varepsilon = 0.03$, where a close three-body encounter drives a brief spike in potential energy; excluding this outlier reduces the maximum to < 3. No drift–verdict correlation is detected; energy-drift distributions, virial-ratio scatter, and convergence curves appear in Appendix A (Figures 14 and 15). The typical per-run drift is below 10^{-3} (median $|\Delta E/E_0|$ across all families); 71 runs ($< 0.1\%$ of direct-isolated runs) exceed 0.1 and 7 exceed 1.0 (see Appendix A.1). Excluding high-drift runs changes no categorical verdict. Per-family median initial virial ratios are $Q_0 \approx 3.3$ for bimodal (super-virial; two unvirialised sub-clusters), 2.0 for Hernquist, 1.8 for Plummer (both mildly super-virial from the approximate velocity sampling), and 0.01 for cold-clumpy (essentially pressureless). Bootstrap CI widths and $|r|$(VelDisp) both stabilise by $\sim$200 replicates, confirming that 500 replicates are sufficient.

PM periodic runs do not permit direct energy bookkeeping because their Hamiltonian differs from the direct-sum Hamiltonian.

Table 1. Conservation diagnostics (direct-isolated model), pooled across all families, N, and ε. Rows show relative energy drift $\Delta E/E_0$, initial virial ratio $Q_0 = 2K_0/|W_0|$, and final virial ratio $Q_f = 2K_f/|W_f|$. Columns show the median and, for energy drift, the single maximum; for virial ratios the third column shows the Min/Max range (the "Max" header is shorthand for this range entry). The maximum drift (~ 17) is a single Hernquist run at $N = 2048$, $\varepsilon = 0.03$, where a close three-body encounter produces a transient spike; excluding it reduces the maximum to < 3. No drift–verdict correlation is detected. PM-periodic runs are excluded.

Diagnostic	Median	Max
Relative energy drift	4.48e-04	1.69e+01
Initial virial ratio	2.023	0.004/5.295
Final virial ratio	2.046	0.279/37.528

5.2. Calibration anchor: bimodal coarse dominance

The bimodal initial condition is the paper's calibration anchor. Figure 2 shows (left) the scatter of initial $\sigma_\rho^2(8)$ against $\Delta C_8^{\text{early}}$ across 500 replicates at $N = 1024$, $\varepsilon = 0.05$, and (right) $|r|$ for CoarseG8 and the best fine observable as a function of softening ε. The coarse predictor achieves $r = 0.984$. The best fine observable in this cell achieves $r = 0.629$. The winner gap (fine − coarse) is -0.357 (CI $[-0.42, -0.30]$, entirely below zero, confirming coarse dominance).

Table 2 shows N-scaling. The coarse predictor is stable across $N \in [256, 2048]$ and the verdict is COARSE in every bimodal cell across all N, ε, and both force models.

Figure 3 illustrates the mechanism, showing the coarse grid overlaid on the particle projection and the $\sigma_\rho^2(8)$–$\Delta C_8^{\text{early}}$ scatter at two particle counts.

5.3. Verdict map

Figure 4 shows verdicts at $N = 1024$ for all families, softening values, and both force models. Concentrated profiles transition from fine at low ε through tie to coarse at larger ε.

Inferential strength is assessed by bootstrap CIs (Figure 5). The bimodal coarse verdict is robustly established (gap CIs exclude zero at all ε). For Hernquist, the fine verdict at $\varepsilon = 0.02$–0.05 is also established; the boundary near $\varepsilon = 0.07$ is unresolved. Plummer fine is established only at $\varepsilon = 0.02$.

Figure 5 shows the continuous effect-size structure underlying these categorical verdicts.

5.4. Time evolution

Figure 6 shows projected-density snapshots for three representative cases: the bimodal coarse-win anchor, the Hernquist fine-win case at $\varepsilon = 0.02$, and the Hernquist coarse-win case at $\varepsilon = 0.10$.

The bimodal sub-clusters merge by the early horizon. The two Hernquist rows appear similar in projected density; the predictive difference between them is kinematic (Figure 8).

5.5. Fine kinematic advantage at small softening

For concentrated profiles at small softening, the local velocity dispersion shows the strongest fine signal in the battery. Table 3 shows VelDisp and kNN-all correlations across all five softening values for Hernquist and Plummer at $N = 1024$. For Hernquist at $\varepsilon = 0.02$: $r_{\text{VD}} = -0.430$ (95% CI $[-0.50, -0.36]$); the kinematic gap CI is $[+0.28, +0.53]$, entirely above zero, confirming that VelDisp outperforms the best coarse predictor at this softening. For Plummer at $\varepsilon = 0.02$: $r_{\text{VD}} = -0.577$

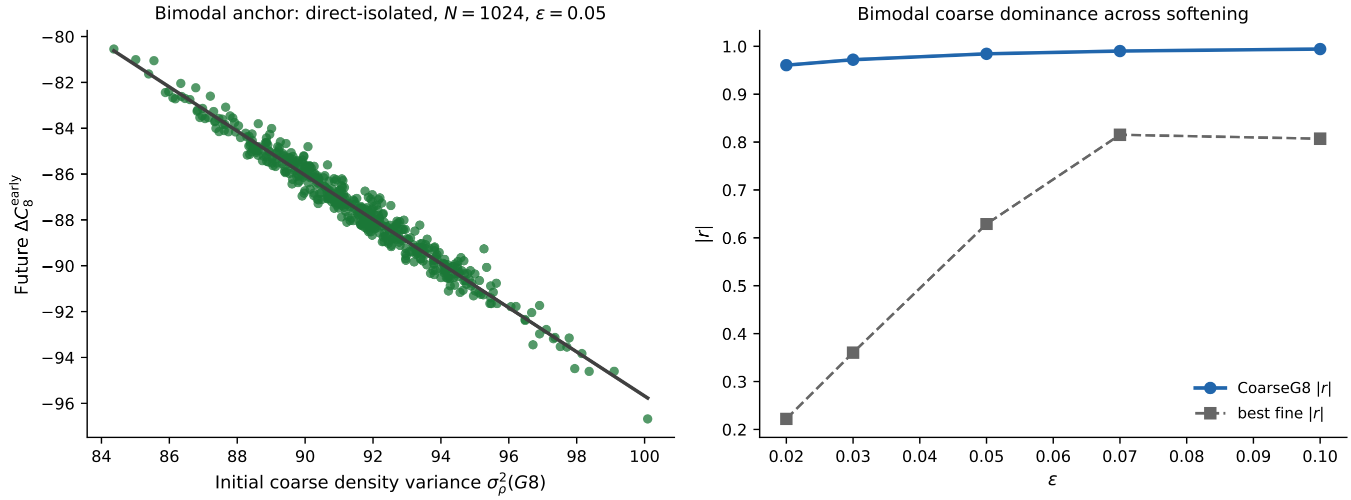

Figure 2. Bimodal calibration anchor: $N = 1024$, $\varepsilon = 0.05$, direct-isolated, 500 replicates. *Left*: initial $\sigma_\rho^2(8)$ vs $\Delta C_8^{\mathrm{early}}$ with least-squares fit. *Right*: $|r|$ for CoarseG8 (blue) and best fine observable (grey) vs softening ε. Coarse dominates at all tested softening values.

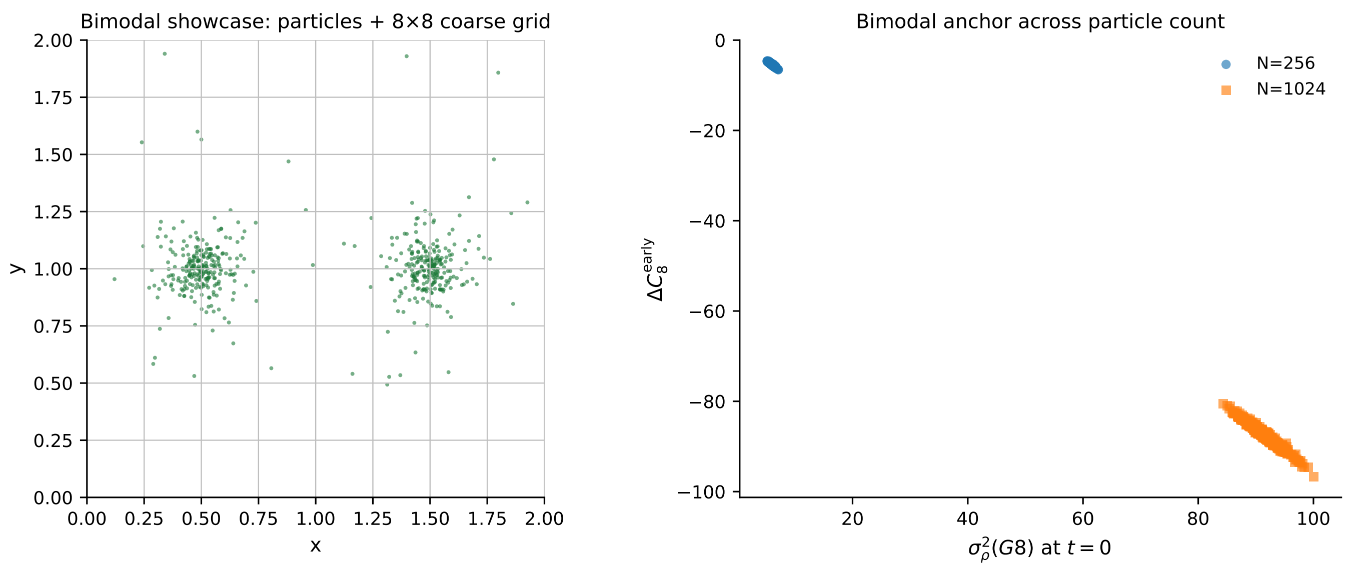

Figure 3. Bimodal mechanism: the inter-cluster density contrast is the predictive object. *Left*: x–y projection of the bimodal IC with projected coarse-grid cells overlaid; cell shade proportional to projected particle count (illustrative; $\sigma_\rho^2(8)$ is computed from the full 3D field). *Right*: $\sigma_\rho^2(8)$ vs $\Delta C_8^{\mathrm{early}}$ at $N = 256$ (circles) and $N = 1024$ (squares), $\varepsilon = 0.05$. The negative association persists across particle count.

(95% CI $[-0.63, -0.52]$), also with a gap CI excluding zero. The fine advantage erodes with increasing ε: Hernquist reaches a tie at $\varepsilon = 0.07$ and coarse dominance at $\varepsilon = 0.10$; Plummer transitions to tie by $\varepsilon = 0.03$ and coarse dominance by $\varepsilon = 0.07$.

Figure 7 shows how VelDisp, kNN-all, and CoarseG8 change with ε across Hernquist and Plummer. For concentrated profiles the VelDisp signal decays with increasing ε, reaching $r_{\mathrm{VD}} = -0.364$ for Hernquist at $\varepsilon = 0.10$.

Figure 8 illustrates the softening dependence: at $\varepsilon = 0.02$ the velocity-dispersion map shows a concentrated hot core; at $\varepsilon = 0.10$ ($\varepsilon/a \approx 0.5$) the cusp dynamics are smoothed out and the kinematic signal disappears.

5.6. What the kinematic advantage is: velocity scale, not local structure

The kinematic advantage is the one regime in this battery where a fine observable beats the coarse predictor, and the observable's definition invites reading it as evi-

Table 2. N-scaling summary for all four IC families. Columns show r_{CG8} at representative softening values and the maximum $|r|$ across fine observables. Rows with $N \leq 2048$ use direct-isolated; $N > 2048$ use PM-periodic (separate experiments). The bimodal verdict is COARSE in every cell; verdicts for other families vary with softening.

IC	N	$\lvert r\rvert_{\mathrm{bc}}(0.02)$	$\lvert r\rvert_{\mathrm{bc}}(0.05)$	$\lvert r\rvert_{\mathrm{bc}}(0.10)$	$\lvert r\rvert_{\mathrm{bf,max}}$	Verdict(0.05)	Model
bimodal	256	+0.923	+0.969	+0.989	0.793	COARSE	direct
bimodal	512	+0.956	+0.982	+0.992	0.821	COARSE	direct
bimodal	1024	+0.961	+0.984	+0.994	0.807	COARSE	direct
bimodal	2048	+0.967	+0.986	+0.995	0.879	COARSE	direct
bimodal	4096	+0.981	+0.981	+0.981	0.834	COARSE	PM
bimodal	8192	+0.985	+0.985	+0.985	0.814	COARSE	PM
bimodal	16384	+0.979	+0.979	+0.979	0.861	COARSE	PM
Hernquist	256	+0.310	+0.142	+0.648	0.519	FINE	direct
Hernquist	512	+0.272	+0.104	+0.665	0.527	FINE	direct
Hernquist	1024	+0.264	+0.176	+0.688	0.582	FINE	direct
Hernquist	2048	+0.270	+0.184	+0.674	0.577	FINE	direct
Hernquist	4096	+0.128	+0.128	+0.128	0.441	FINE	PM
Hernquist	8192	+0.183	+0.183	+0.183	0.386	TIE	PM
Hernquist	16384	+0.314	+0.314	+0.314	0.432	TIE	PM
Plummer	256	+0.321	+0.571	+0.854	0.690	TIE	direct
Plummer	512	+0.379	+0.590	+0.858	0.695	TIE	direct
Plummer	1024	+0.382	+0.603	+0.871	0.736	TIE	direct
Plummer	2048	+0.438	+0.625	+0.868	0.629	TIE	direct
Plummer	4096	+0.465	+0.465	+0.465	0.562	TIE	PM
Plummer	8192	+0.527	+0.527	+0.527	0.513	TIE	PM
Plummer	16384	+0.428	+0.428	+0.428	0.565	TIE	PM
cold-clumpy	256	+0.316	+0.277	+0.327	0.420	TIE	direct
cold-clumpy	512	+0.277	+0.267	+0.266	0.467	TIE	direct
cold-clumpy	1024	+0.261	+0.245	+0.239	0.457	TIE	direct
cold-clumpy	2048	+0.239	+0.223	+0.227	0.458	TIE	direct
cold-clumpy	4096	+0.229	+0.229	+0.229	0.462	TIE	PM
cold-clumpy	8192	+0.218	+0.218	+0.218	0.447	TIE	PM
cold-clumpy	16384	+0.233	+0.233	+0.233	0.440	TIE	PM

dence that *local* phase-space structure carries predictive information. A preregistered re-analysis shows this reading is largely incorrect: the advantage is predominantly a *bulk velocity-scale* (temperature) effect.

Across the per-realisation initial states, VelDisp is 0.95–0.96 correlated with the total initial kinetic energy: as a single number per realisation, the local-dispersion average tracks the realisation's overall velocity scale far more than its spatial velocity texture. To separate these, we score four nested models by held-out R^2 (RidgeCV, 200 repeated train/test splits, predicting the absolute future $\sigma_\rho^2(8)$ so the shared-baseline term cannot inflate any increment, with the initial coarse field in every baseline): a coarse-positional baseline X; X+VelDisp; X+GlobalVscale, where GlobalVscale is VelDisp with locality removed (the whole-system velocity dispersion, $\sqrt{2K_0}$); and X+VelDisp+GlobalVscale. The fraction of VelDisp's incremental advantage recovered by the single global scalar, and the fraction unique to VelDisp beyond it, follow directly from the R^2 increments with paired-bootstrap CIs.

The result is uniform across families. The global velocity scale recovers 90% ($[0.70, 1.03]$), 83% ($[0.74, 0.91]$), and 81% ($[0.64, 0.93]$) of VelDisp's held-out advantage for

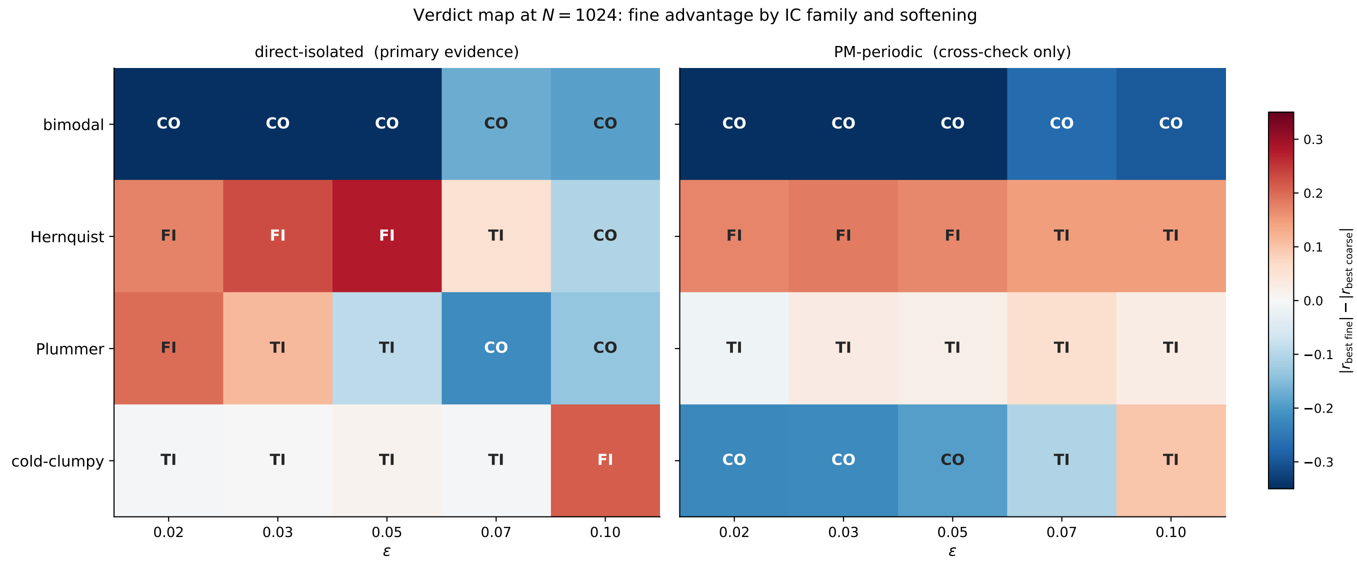

Figure 4. Verdict map at $N = 1024$, primary target $\Delta C_8^{\text{early}}$. Each cell shows the categorical verdict (CO = COARSE, FI = FINE, TI = tie). Blue: COARSE. Green: FINE. Grey: tie. *Left*: direct-isolated. *Right*: PM-periodic. Bimodal is COARSE in all cells of both panels. Continuous effect sizes are shown in Figure 5.

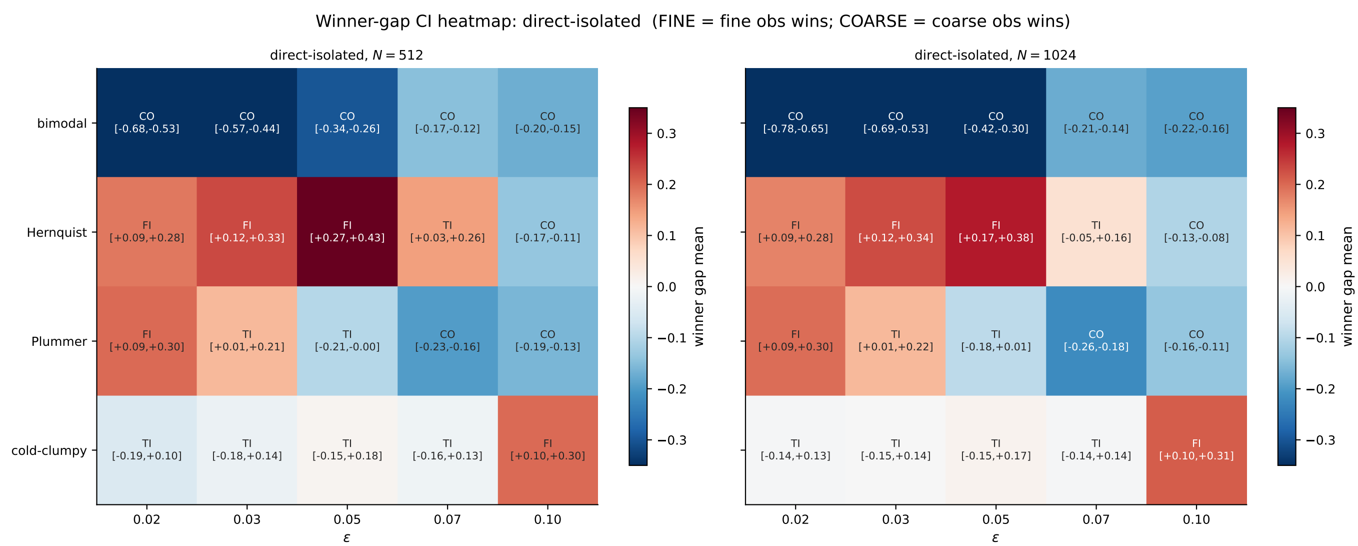

Figure 5. Winner-gap effect size and 95% bootstrap CI for each parameter cell at $N = 1024$. Colour encodes the point-estimate winner gap $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$ (negative = coarse dominance); error bars show the bootstrap CI. Bimodal cells cluster at large negative gaps with CIs entirely below zero. Boundary cells (Hernquist at $\varepsilon = 0.07$, Plummer at $\varepsilon = 0.03$–0.05) have CIs spanning zero.

Hernquist ($\varepsilon = 0.02$), Plummer ($\varepsilon = 0.02$), and bimodal respectively, while VelDisp's unique increment beyond the global scale is only 9%, 19%, and 19% (CI excludes zero only for the core and merger cells). A locality ladder (neighbourhood size $k = 16, 64, 256, N$) is not flat but peaks at large k and falls toward the fully global limit, indicating a modest intermediate-scale component superposed on a velocity-scale-dominated signal. The kinematic advantage is therefore ∼85% a temperature effect: realisations that are hotter overall evolve their grid-scale clustering differently, and local velocity dispersion is mostly an estimator of that global scale. This does not overturn the observable-class verdict—VelDisp remains the fine observable that beats coarse—but it corrects *why*: the channel is the system's velocity scale, with only a small genuinely local phase-space contribution.

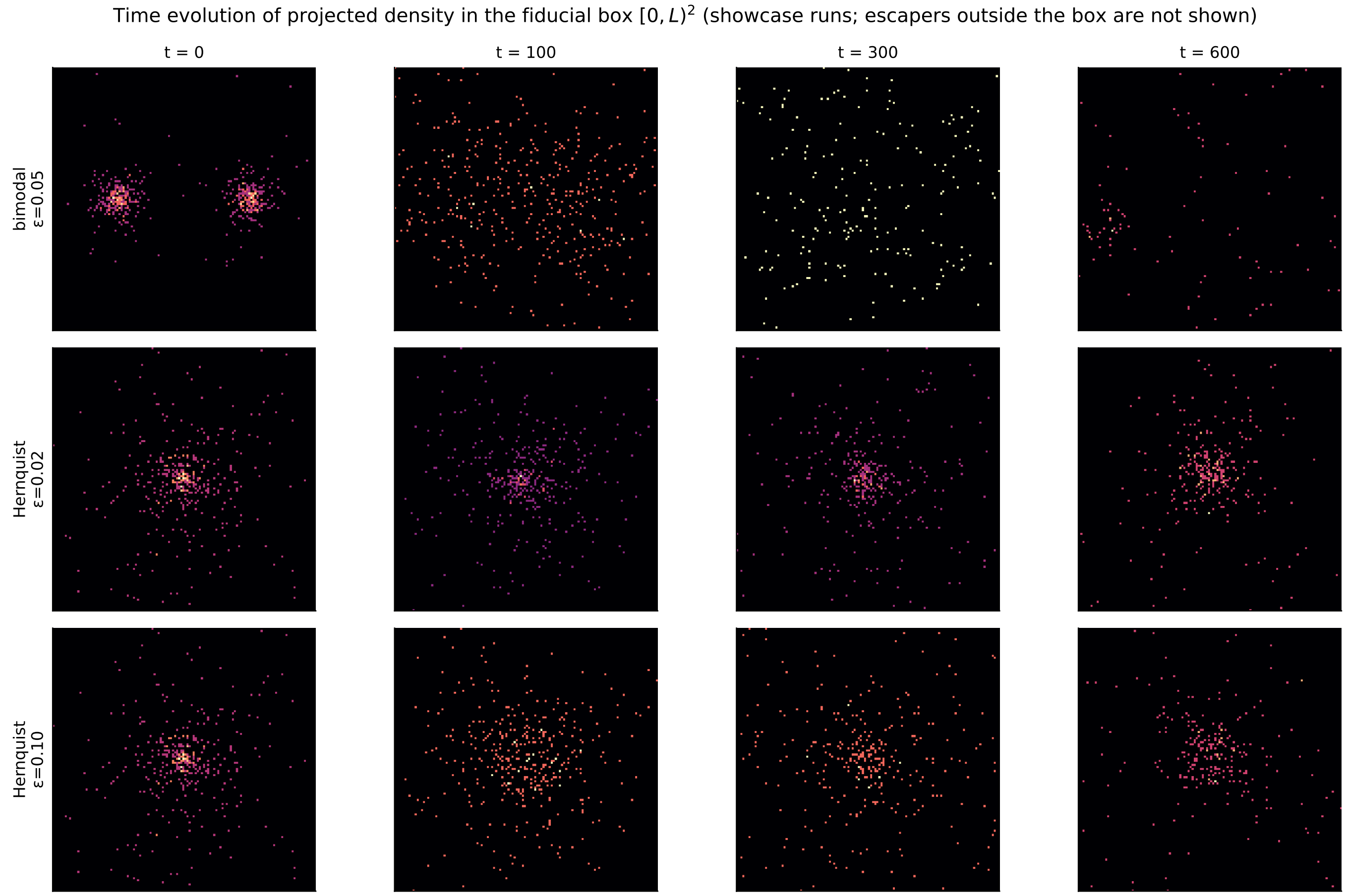

Figure 6. Time evolution: projected density (log scale, $N = 512$, direct-isolated). *Rows*: bimodal ($\varepsilon = 0.05$); Hernquist ($\varepsilon = 0.02$); Hernquist ($\varepsilon = 0.10$). *Columns*: $t = 0$, early, mid, final. The two Hernquist rows appear visually similar in projected density; the predictive difference between them is kinematic, not spatial (see Figure 8).

Table 3. VelDisp and kNN-all correlations across softening for Hernquist and Plummer at $N = 1024$, direct-isolated. VelDisp outperforms the coarse predictor at low ε; kNN-all does not. Overall winner-gap CIs (finest vs. coarse class) exclude zero for Hernquist at $\varepsilon \leq 0.05$ and Plummer at $\varepsilon = 0.02$. Kinematic-class CIs (VelDisp vs. CoarseG8) additionally exclude zero at Hernquist $\varepsilon = 0.07$ and Plummer $\varepsilon = 0.03$–0.05. Note: the overall winner-gap CI uses the *best fine observable across all classes* vs. the best coarse observable; the kinematic-class CI is the narrower pairwise comparison of VelDisp alone against CoarseG8. A cell can show a tie or coarse verdict overall while the VelDisp-vs-coarse pairwise CI excludes zero, because kNN-all and other positional fine observables pull the best-fine envelope down.

IC	ϵ	r_{VelDisp}	r_{kNN}	Pos. gap CI	Kin. gap CI	Verdict
Hernquist	0.02	-0.430	+0.370	[+0.08, +0.19]	[+0.28, +0.53]	FINE
Hernquist	0.03	-0.437	+0.332	[+0.07, +0.21]	[+0.36, +0.59]	FINE
Hernquist	0.05	-0.459	+0.177	[-0.05, +0.19]	[+0.45, +0.66]	FINE
Hernquist	0.07	-0.434	-0.162	[-0.16, -0.08]	[+0.19, +0.40]	TIE
Hernquist	0.10	-0.364	-0.582	[-0.09, -0.04]	[-0.09, +0.10]	COARSE
Plummer	0.02	-0.577	+0.070	[-0.37, -0.13]	[+0.41, +0.61]	FINE
Plummer	0.03	-0.566	+0.005	[-0.39, -0.23]	[+0.28, +0.49]	TIE
Plummer	0.05	-0.511	-0.162	[-0.37, -0.27]	[+0.09, +0.29]	TIE
Plummer	0.07	-0.431	-0.499	[-0.23, -0.16]	[-0.09, +0.09]	COARSE
Plummer	0.10	-0.318	-0.736	[-0.14, -0.08]	[-0.26, -0.09]	COARSE

5.7. The softening transition

Figure 9 shows for all four families how the winner gap changes with ε.

Bimodal remains strongly negative at all ε because the dynamically decisive scale ($\sim 0.5\,L$) far exceeds the

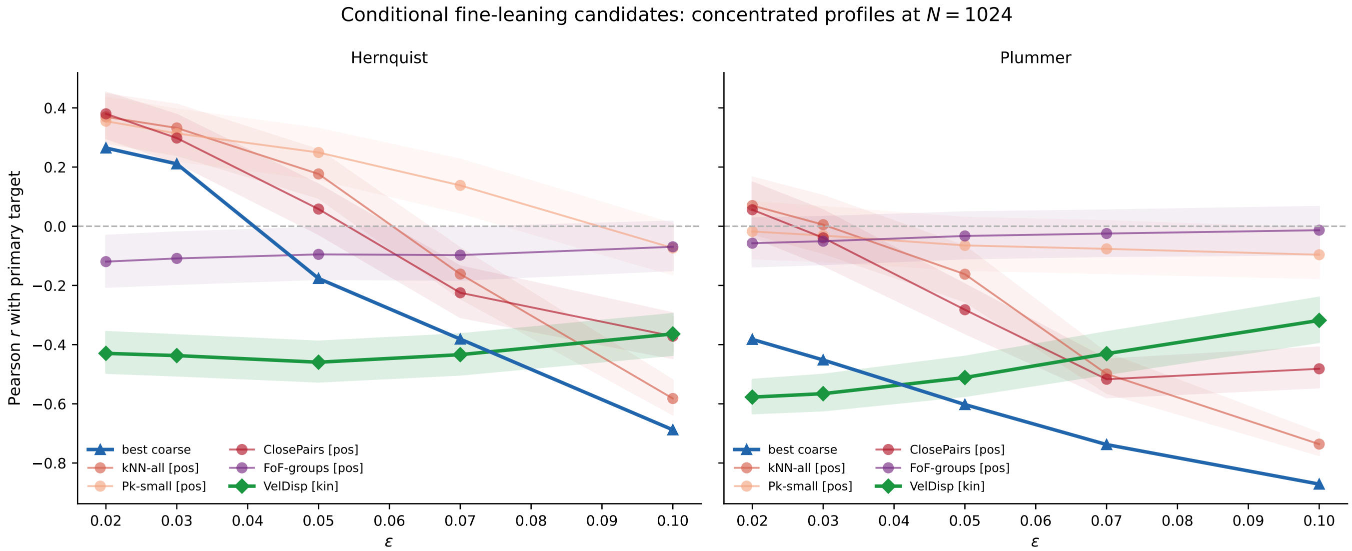

Figure 7. Observable class vs softening: $N = 1024$, direct-isolated. All series use solid lines, distinguished by colour and marker: blue triangles = best coarse predictor; circles = positional fine observables (one per observable, semi-transparent); diamonds = VelDisp (kinematic fine). Shaded bands: 95% bootstrap CI. VelDisp outperforms the coarse predictor at low ε for concentrated profiles; the advantage weakens with increasing ε. Positional fine observables rarely outperform the coarse predictor; the kinematic class drives the fine advantage.

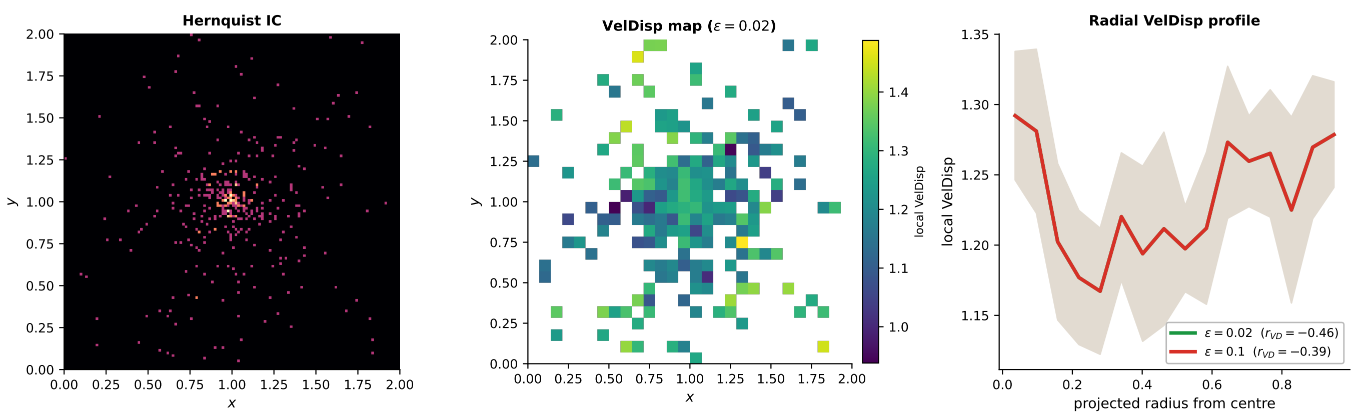

Figure 8. VelDisp mechanism: Hernquist profile at $N = 512$. *Left*: projected particle density. *Centre*: local velocity-dispersion map at $\varepsilon = 0.02$. *Right*: radial VelDisp profiles at $\varepsilon = 0.02$ (green) and $\varepsilon = 0.10$ (red), with $r(\text{VelDisp}, \Delta C_8^{\text{early}})$ annotated. The kinematic hot core is visible at small softening; at large softening the profile flattens and the predictive correlation weakens.

softening length. For Hernquist the winner gap crosses zero near $\varepsilon = 0.07$; Plummer crosses earlier. The relevant ratio is ε/a: the cusp is partially resolved at $\varepsilon/a = 0.10$ and smoothed out at $\varepsilon/a = 0.50$.

Figure 10 plots the winner gap against ε/a. The Hernquist zero-crossing occurs between $\varepsilon/a \approx 0.35$ and 0.50 ($\varepsilon = 0.07$ is a tie; $\varepsilon = 0.10$ is COARSE); Plummer crosses earlier. A finer ε/a sweep would locate the boundary more precisely.

5.8. N-scaling

Figure 11 shows r_{CG8} and the best fine predictor vs N for all four families, at each ε value.

The bimodal verdict is stable across N at all ε. For concentrated profiles, VelDisp improves modestly with N and the fine advantage at small ε persists; no verdict reversal occurs.

5.9. Force-model comparison

Figure 12 shows r_{CG8} vs ε for both model classes at $N = 1024$.

The PM-periodic model serves as a robustness cross-check on the family-level verdict structure rather than co-equal evidence for the ε-dependent transition (which is a direct-isolated phenomenon). Both model classes agree on the bimodal categorical verdict at all ε. For

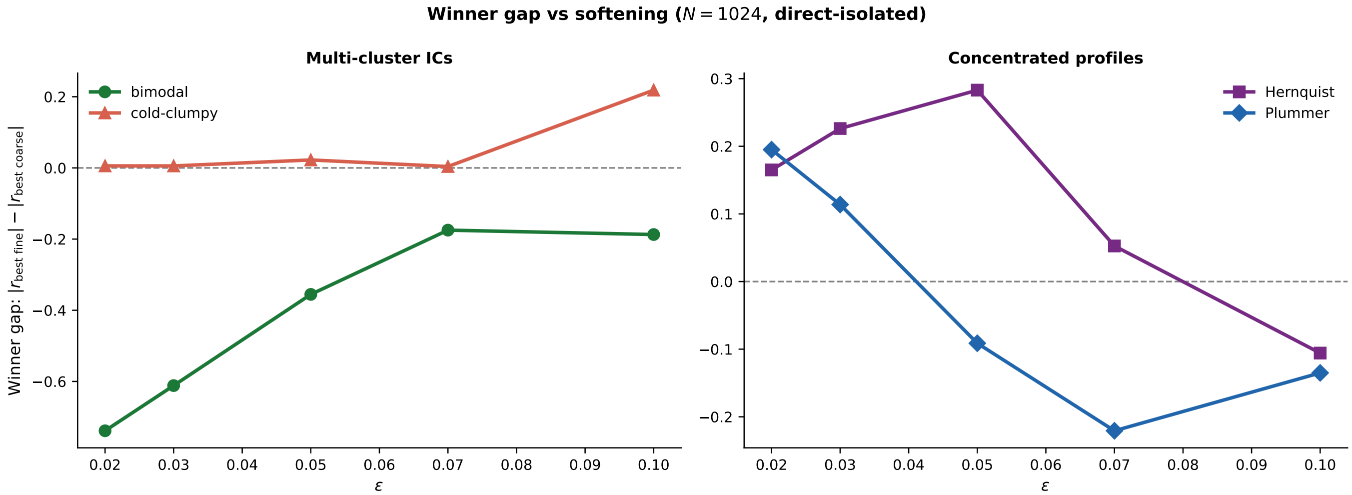

Figure 9. Winner gap vs softening: $N = 1024$, direct-isolated. Winner gap $= |r_{\text{best fine}}| - |r_{\text{best coarse}}|$ (positive = fine advantage, negative = coarse dominance). *Left*: multi-cluster ICs (bimodal, cold-clumpy); bimodal is strongly negative at all ε. *Right*: concentrated profiles (Hernquist, Plummer); Hernquist transitions from FINE to TIE between $\varepsilon = 0.05$ and 0.07, reaching COARSE by $\varepsilon = 0.10$; Plummer transitions earlier.

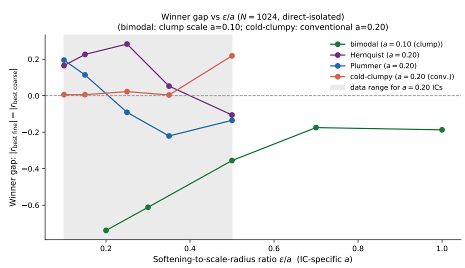

Figure 10. Verdict boundary in ε/a: winner gap ($|r_{\text{best fine}}| - |r_{\text{best coarse}}|$) vs the softening-to-scale-radius ratio, $N = 1024$, direct-isolated. Each point is annotated with the verdict (C = COARSE, F = FINE, T = tie). Bimodal is negative at all tested ε/a. Hernquist and Plummer cross zero between $\varepsilon/a \approx 0.35$ and 0.50 (shaded zone), transitioning from fine to coarse dominance.

concentrated profiles the PM rows are nearly ε-invariant (force resolution set by mesh, not ε), so the direct-isolated results carry the weight of the softening-transition evidence; PM agreement on categorical verdicts provides robustness confirmation (Springel 2005; Dehnen & Read 2011). The cold-clumpy family shows a verdict discrepancy between solvers (TIE in direct-isolated, COARSE in some PM cells): this likely reflects the PM grid's insensitivity to sub-grid clump structure, which the direct solver partially resolves. This discrepancy does not affect the paper's main claims, which rest on bimodal and concentrated-profile cells where both solvers agree.

5.10. Observable-class dependence

The main verdict table (Table 4) summarises verdicts across both force-model classes, all initial-condition families, and all softening values at $N = 1024$.

Three patterns hold: (i) positional fine observables rarely outperform the coarse predictor (the positional-class gap CI excludes zero in only 10 of 80 direct-isolated cells, mostly Hernquist at low ε; the kinematic class dominates the fine advantage); (ii) FoF group count never outperforms the coarse predictor (it may appear as "best fine" in some cells of Table 4, meaning only that it is the least-weak fine observable, not that it outperforms coarse); (iii) VelDisp is the only fine observable whose gap CIs exclude zero, and only for concentrated profiles at small ε. The cold-clumpy family shows a single anomalous FINE verdict at $\varepsilon = 0.10$ (Table 4), driven by the ClosePairs observable. At large softening, the linking radius $4\varepsilon = 0.40$ approaches the inter-clump separation, making ClosePairs a proxy for inter-clump connectivity rather than true fine-scale structure. This cell does not reflect a physical fine-scale advantage.

5.11. Summary matrix

Figure 13 shows r for all observable class $\times$ ε combinations as a colour matrix.

The class-level gap distribution (Appendix A, Figure 17) confirms this structure: VelDisp is the only observable class whose gap distribution extends above zero for concentrated profiles at low ε; positional fine classes remain at or below zero in all tested cells.

6. DISCUSSION

6.1. Physical interpretation: bimodal case

In a two-cluster merger the dominant dynamical variable is the inter-cluster density contrast, encoded in $\sigma^2_\rho(8)$ at grid scale $L/8 \approx 0.25$. The merger timescale

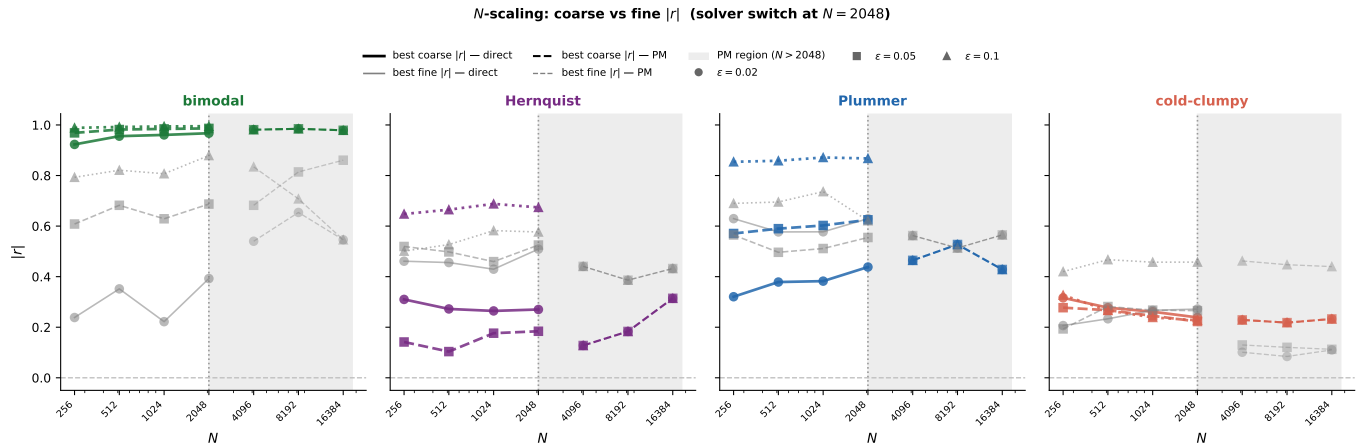

Figure 11. N-scaling at three softening values, direct-isolated. Each panel: one IC family. Coloured lines: r_{CG8} (solid/dashed/dotted for $\varepsilon = 0.02/0.05/0.10$). Grey: best fine predictor. Bimodal is stable near -0.98 at all N and ε.

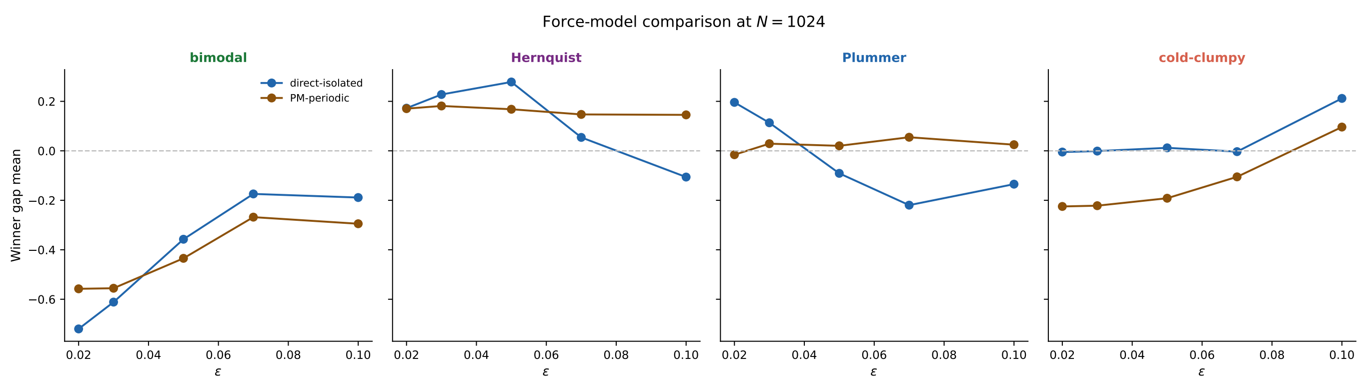

Figure 12. Force-model comparison: r_{CG8} vs ε, $N = 1024$. Blue: direct-isolated. Brown: PM-periodic. Shaded: 95% CI. Both models agree on the bimodal verdict. Concentrated profiles show quantitative differences: direct-isolated exhibits a clear ε-dependent transition while PM-periodic rows are nearly ε-invariant (force resolution is set by the mesh, not by ε; ClosePairs is the only exception because its threshold 4ε changes with ε by construction). PM-periodic ε values therefore scan the ClosePairs physical scale rather than the force resolution; differences from direct-isolated trends in other observables reflect this solver difference.

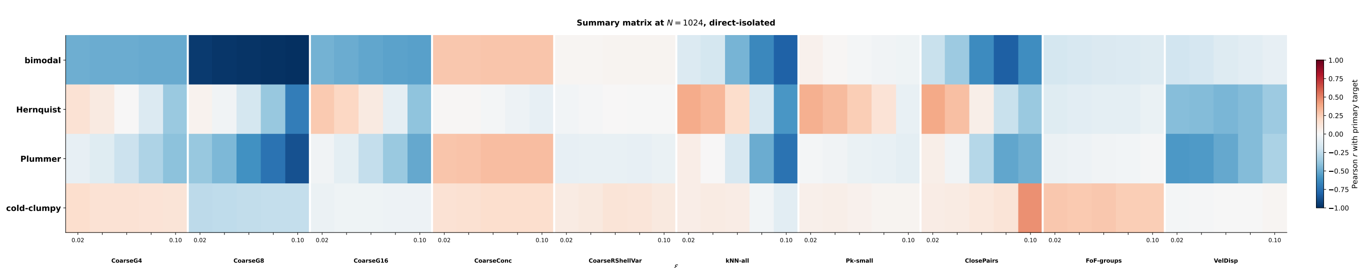

Figure 13. Summary matrix: Pearson r with $\Delta C_8^{\mathrm{early}}$ by IC family (rows), observable class (groups), and ε (within-group columns), at $N = 1024$, direct-isolated. The coarse group (leftmost) shows strong negative r for bimodal at all ε. VelDisp (fine kinematic) shows the strongest fine signal for Hernquist and Plummer at $\varepsilon = 0.02$. Positional fine observables rarely outperform coarse (10 of 80 direct-isolated cells); the kinematic class drives the fine advantage.

$t_{\mathrm{merge}} \sim d/v_{\mathrm{rel}}$ depends on large-scale quantities; realisations with higher initial contrast merge faster and produce more clustering growth by the early horizon. Sub-cluster internal structure is dynamically decoupled at early times, explaining why positional fine observables carry no additional predictive power. This is the controlled analogue of hierarchical halo merging, where the large-scale overdensity at the virial-radius scale—not

Table 4. Verdict summary at $N = 1024$, both force-model classes, all initial-condition families, and $\varepsilon \in \{0.02, 0.03, 0.05, 0.07, 0.10\}$. Gap column shows $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$.

Model	IC	ϵ	Best coarse	$\lvert r\rvert_{\text{bc}}$	Best fine	$\lvert r\rvert_{\text{bf}}$	Gap mean	Gap CI	Verdict	n_{lw}
direct-isolated	bimodal	0.02	CoarseG8	+0.961	ClosePairs	+0.222	-0.720	[-0.78, -0.65]	COARSE	500
direct-isolated	bimodal	0.03	CoarseG8	+0.972	ClosePairs	+0.360	-0.611	[-0.69, -0.53]	COARSE	500
direct-isolated	bimodal	0.05	CoarseG8	+0.984	ClosePairs	+0.629	-0.357	[-0.42, -0.30]	COARSE	500
direct-isolated	bimodal	0.07	CoarseG8	+0.990	ClosePairs	+0.815	-0.174	[-0.21, -0.14]	COARSE	500
direct-isolated	bimodal	0.10	CoarseG8	+0.994	kNN-all	+0.807	-0.189	[-0.22, -0.16]	COARSE	500
direct-isolated	Hernquist	0.02	CoarseG16	+0.264	VelDisp	+0.430	+0.172	[+0.09, +0.28]	FINE	500
direct-isolated	Hernquist	0.03	CoarseG16	+0.211	VelDisp	+0.437	+0.228	[+0.12, +0.34]	FINE	500
direct-isolated	Hernquist	0.05	CoarseG8	+0.176	VelDisp	+0.459	+0.278	[+0.17, +0.38]	FINE	500
direct-isolated	Hernquist	0.07	CoarseG8	+0.381	VelDisp	+0.434	+0.054	[-0.05, +0.16]	TIE	500
direct-isolated	Hernquist	0.10	CoarseG8	+0.688	kNN-all	+0.582	-0.106	[-0.13, -0.08]	COARSE	500
direct-isolated	Plummer	0.02	CoarseG8	+0.382	VelDisp	+0.577	+0.196	[+0.09, +0.30]	FINE	500
direct-isolated	Plummer	0.03	CoarseG8	+0.452	VelDisp	+0.566	+0.113	[+0.01, +0.22]	TIE	500
direct-isolated	Plummer	0.05	CoarseG8	+0.603	VelDisp	+0.511	-0.091	[-0.18, +0.01]	TIE	500
direct-isolated	Plummer	0.07	CoarseG8	+0.738	ClosePairs	+0.517	-0.220	[-0.26, -0.18]	COARSE	500
direct-isolated	Plummer	0.10	CoarseG8	+0.871	kNN-all	+0.736	-0.135	[-0.16, -0.11]	COARSE	500
direct-isolated	cold-clumpy	0.02	CoarseG8	+0.261	FoF-groups	+0.267	-0.005	[-0.14, +0.13]	TIE	500
direct-isolated	cold-clumpy	0.03	CoarseG8	+0.257	FoF-groups	+0.262	-0.001	[-0.15, +0.14]	TIE	500
direct-isolated	cold-clumpy	0.05	CoarseG8	+0.245	FoF-groups	+0.268	+0.012	[-0.15, +0.17]	TIE	500
direct-isolated	cold-clumpy	0.07	CoarseG8	+0.239	FoF-groups	+0.243	-0.003	[-0.14, +0.14]	TIE	500
direct-isolated	cold-clumpy	0.10	CoarseG8	+0.239	ClosePairs	+0.457	+0.212	[+0.10, +0.31]	FINE	500
PM-periodic	bimodal	0.02	CoarseG8	+0.978	FoF-groups	+0.568	-0.558	[-0.62, -0.50]	COARSE	500
PM-periodic	bimodal	0.03	CoarseG8	+0.978	FoF-groups	+0.568	-0.555	[-0.62, -0.50]	COARSE	500
PM-periodic	bimodal	0.05	CoarseG8	+0.978	ClosePairs	+0.612	-0.435	[-0.49, -0.38]	COARSE	500
PM-periodic	bimodal	0.07	CoarseG8	+0.978	ClosePairs	+0.793	-0.268	[-0.30, -0.23]	COARSE	500
PM-periodic	bimodal	0.10	CoarseG8	+0.978	kNN-all	+0.784	-0.295	[-0.33, -0.26]	COARSE	500
PM-periodic	Hernquist	0.02	CoarseG16	+0.239	VelDisp	+0.453	+0.170	[+0.07, +0.28]	FINE	500
PM-periodic	Hernquist	0.03	CoarseG16	+0.239	VelDisp	+0.453	+0.181	[+0.07, +0.29]	FINE	500
PM-periodic	Hernquist	0.05	CoarseG16	+0.239	VelDisp	+0.453	+0.168	[+0.05, +0.28]	FINE	500
PM-periodic	Hernquist	0.07	CoarseG16	+0.239	VelDisp	+0.453	+0.147	[+0.03, +0.27]	TIE	500
PM-periodic	Hernquist	0.10	CoarseG16	+0.239	VelDisp	+0.453	+0.145	[+0.03, +0.27]	TIE	500
PM-periodic	Plummer	0.02	CoarseG8	+0.451	VelDisp	+0.563	-0.016	[-0.11, +0.09]	TIE	500
PM-periodic	Plummer	0.03	CoarseG8	+0.451	VelDisp	+0.563	+0.029	[-0.07, +0.13]	TIE	500
PM-periodic	Plummer	0.05	CoarseG8	+0.451	VelDisp	+0.563	+0.020	[-0.09, +0.13]	TIE	500
PM-periodic	Plummer	0.07	CoarseG8	+0.451	VelDisp	+0.563	+0.055	[-0.04, +0.15]	TIE	500
PM-periodic	Plummer	0.10	CoarseG8	+0.451	VelDisp	+0.563	+0.025	[-0.07, +0.13]	TIE	500
PM-periodic	cold-clumpy	0.02	CoarseG8	+0.248	ClosePairs	+0.037	-0.225	[-0.31, -0.12]	COARSE	500
PM-periodic	cold-clumpy	0.03	CoarseG8	+0.248	ClosePairs	+0.038	-0.222	[-0.31, -0.12]	COARSE	500
PM-periodic	cold-clumpy	0.05	CoarseG8	+0.248	ClosePairs	+0.086	-0.192	[-0.31, -0.06]	COARSE	500
PM-periodic	cold-clumpy	0.07	CoarseG8	+0.248	ClosePairs	+0.191	-0.105	[-0.25, +0.03]	TIE	500
PM-periodic	cold-clumpy	0.10	CoarseG8	+0.248	ClosePairs	+0.449	+0.096	[-0.04, +0.21]	TIE	500

internal substructure—governs the merger rate (White 1996).

6.2. **Physical interpretation: concentrated profiles**

A plausible interpretation is that local velocity dispersion captures support against near-term collapse in a way unavailable to purely positional summaries: regions with higher initial σ_v^{loc} have more virial support and resist collapse, producing a negative correlation with future clustering growth. Positional fine observables cannot detect this because the equilibrium density profile is smooth above the softening scale; the fine advantage is observable-class-specific, requiring *velocity* access. The commonality re-analysis of Section 5.6 sharpens this interpretation: the velocity channel that carries the advantage is predominantly the *global* velocity scale (overall virial temperature of the realisation), not local phase-space texture, so "support against near-term collapse" is better read at the system scale than at the neighbourhood scale. It should be noted that even in the best-case cells (Hernquist, $\varepsilon \leq 0.05$), $|r_{\text{VD}}| \approx 0.45$–$0.55$, implying $R^2 \approx 0.18$–0.28 at $N = 1024$: VelDisp is the *best available* predictor in these regimes but still explains only 20–30% of the variance in $\Delta C_8^{\text{early}}$. Much of the cusp evolution remains stochastic or depends on degrees of freedom not captured by the scalar initial observables considered here.

Increasing ε smooths the potential over scales comparable to the cusp radius a. The transition from fine to coarse dominance lies between $\varepsilon/a \approx 0.35$ and 0.50 for Hernquist, and at lower ε/a for Plummer. The five-point grid resolves the direction but not the precise boundary.

This transition connects to the violent-relaxation picture of Lynden-Bell (1967): when ε is small enough to resolve the cusp, rapid potential fluctuations during initial collapse mix phase space while preserving kinematic substructure at the resolved scale, leaving VelDisp as a predictive residue of incomplete memory erasure (Chavanis 2002). This is consistent with the finding that merger remnants retain kinematic signatures (shells, streams, velocity anisotropy) long after spatial relaxation (Hernquist & Spergel 1992): the VelDisp advantage we measure is the predictive counterpart of this incomplete phase-space mixing. The battery supports a consistent mechanistic interpretation: when ε exceeds the cusp scale radius, the potential is smoothed over the region that carries kinematic substructure; phase-space memory becomes inaccessible, and only the coarse density contrast retains predictive value. For dark matter haloes with NFW-like cusps (Navarro et al. 1996, 1997), the transition range $\varepsilon/a \approx 0.35$–$0.50$ maps onto scales comparable to the convergence radius identified by Power et al. (2003), but any direct connection would require dedicated investigation with realistic populations and is not claimed here.

Radial-distance controls and permutation-null shuffles confirm that the main predictive signals are not trivially explained by geometry or statistical artefacts; these diagnostic panels appear in Appendix A (Figure 16).

6.3. Astrophysical implications

The bimodal coarse-dominance result is consistent with the hierarchical picture in which large-scale overdensity governs merger timing (White 1996): the inter-cluster density contrast at scale $L/8$ plays a role analogous to the overdensity that drives dynamical friction in galaxy mergers.

The ε/a transition for concentrated profiles reflects resolution-dependent access to cusp kinematics within this controlled setting. The transition range $\varepsilon/a \approx 0.35$–$0.50$ is a feature of these idealised families; it is not a prescriptive design rule and should not be extrapolated to cosmological populations, which include tidal fields, baryonic processes, and hierarchical assembly.

6.4. What is and is not earned

Earned within this battery:

- Robust coarse dominance for bimodal ICs across all tested N, ε, and both force models.
- Positional fine observables rarely outperform the coarse predictor (10 of 80 direct-isolated cells, concentrated at Hernquist low-ε and cold-clumpy $\varepsilon = 0.10$); the kinematic class dominates the fine advantage wherever it appears.
- Fine kinematic (VelDisp) advantage for Hernquist at $\varepsilon \leq 0.05$ and Plummer at $\varepsilon = 0.02$ (gap CIs exclude zero).
- The kinematic advantage is predominantly a bulk velocity-scale (temperature) effect: a single global velocity-scale scalar recovers 81–90% of it, with only a small local residual (Section 5.6).
- ε-dependent transition from fine to coarse dominance for concentrated profiles, governed in part by ε/a.
- All load-bearing claims survive direct pairwise comparison ($|r_{\mathrm{VelDisp}}| > |r_{\mathrm{CoarseG8}}|$ in the same cells) and removal of the null-bias correction.

Unresolved:

- Hernquist at $\varepsilon = 0.07$ and Plummer at $\varepsilon = 0.03$–0.05: gap CIs span zero.
- Exact transition boundary in ε/a.
- Whether results generalise beyond these four IC families and this target.

Not earned:

- Universal ranking across self-gravitating systems.
- Sharp critical transition in ε/a.
- Causal mechanism (this study measures correlation, not causation).
- Cosmological relevance (no expansion, tidal fields, or baryons).
- Family-stable law (43 of 280 cells with ≥ 100 listwise-complete replicates are family-stable).

All main verdicts are reproduced when the coarse final state $\sigma_\rho^2(8)|_{t=t_e}$ replaces the difference $\Delta C_8^{\mathrm{early}}$ as the target, ruling out shared-baseline deflation (Section 4.2).

6.5. Future work

Several extensions would strengthen and broaden these results. First, a denser ε/a sweep in the transition region ($\varepsilon/a \in [0.30, 0.55]$) for Hernquist and Plummer would sharpen the boundary location; this is computationally tractable within the existing framework. Second, testing whether the VelDisp advantage persists at the intermediate horizon (step 300, $t = 1.5$) would clarify whether the fine-kinematic signal is a transient feature of the initial violent-relaxation phase or a persistent predictive advantage. Third, the prediction target is a single scalar ($\Delta C_8^{\rm early}$); fine observables might carry more information for spatially resolved targets (e.g. radial profile shape) or longer prediction horizons. Finally, extending the battery to cosmological initial conditions—with tidal fields, hierarchical assembly, and baryonic processes—would test whether the family-dependent verdict structure observed here survives in more realistic settings.

7. CONCLUSION

This battery of 140000 simulations across 7 families, 2 force models, and five softening values yields three results.

Primary: coarse density variance robustly dominates for merging binary clusters across every tested cell ($r = 0.984$ at $\varepsilon = 0.05$; winner-gap CIs entirely below zero at all N, ε, and both force models).

Secondary: local velocity dispersion outperforms the coarse predictor for concentrated cusp profiles in a restricted low-softening regime (Hernquist $\varepsilon \leq 0.05$, Plummer $\varepsilon = 0.02$; gap CIs exclude zero), with a transition to coarse dominance by $\varepsilon = 0.10$.

Negative: positional fine-scale observables rarely outperform the coarse predictor (the positional-class gap CI excludes zero in only 10 of 80 direct-isolated cells, concentrated at Hernquist low-ε and cold-clumpy $\varepsilon = 0.10$). The dominant fine advantage is kinematic, not positional, and requires three conditions simultaneously: a concentrated-cusp family, small ε/a, and phase-space access.

No universal ranking is supported. Predictive advantage is conditional on the initial-condition family, the observable class, and the softening regime—not a general property of self-gravitating systems.

All results are specific to the scalar target $\Delta C_8^{\rm early}$ and the stylised families studied here; the paper is fundamentally an observable-class comparison under a fixed target choice, not a broader characterisation of predictive structure in gravity.

ACKNOWLEDGMENTS

Acknowledgments withheld for double-blind review.

Software: numpy (Harris et al. 2020), scipy (Virtanen et al. 2020), matplotlib (Hunter 2007)

DATA AND CODE AVAILABILITY

All simulation code, analysis pipeline, figure generation, and regression tests are available in a repository withheld for double-blind review; the URL and Zenodo DOI will be provided in the accepted version. The repository includes: `nbody_stress.py` (battery runner and statistical analysis), `nbody_paper.py` (figure and table generation), `nbody_3d.py` (force models and integrators), `test_regression.py` (34 regression tests), and `build.sh` (reproducible build script). The full 140000-run battery CSV ($\sim$ 53 MB) is not included in the repository but can be regenerated from scratch using `python nbody_paper.py --workers 30` (approximately 28 hours on 32 vCPUs) or reproduced via `--resume` for incremental runs. Pre-computed figures and tables are included in `outputs/`.

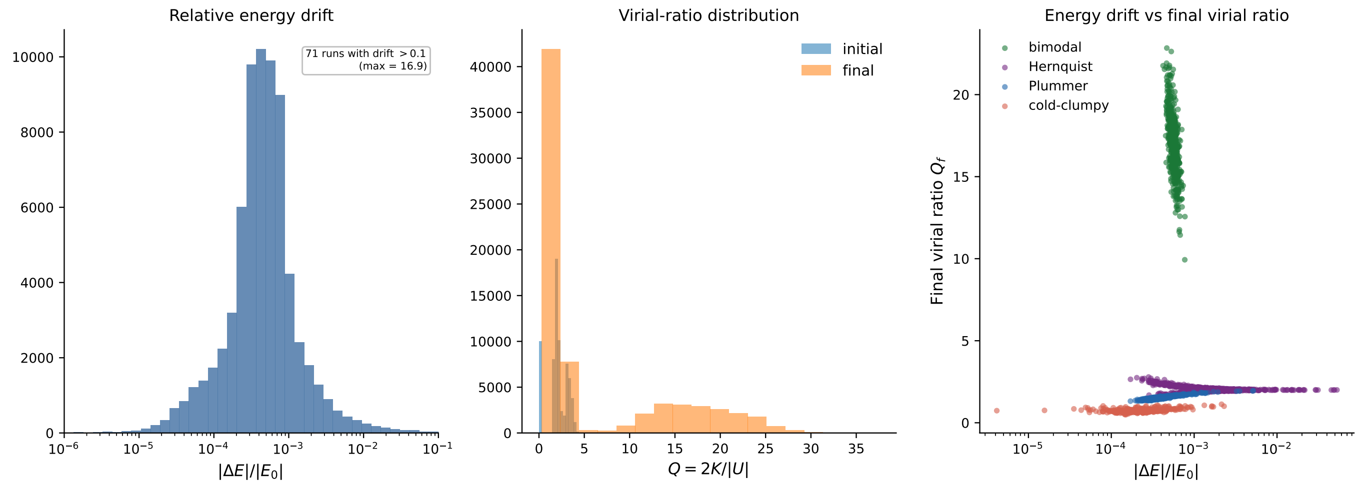

Figure 14. *Left*: distribution of $\log_{10}(\Delta E/E_0)$ by IC family (direct-isolated). *Centre*: initial virial ratio $Q_0 = 2K/|W|$ by family; dashed line marks virial equilibrium $Q = 1$. *Right*: energy drift vs final virial ratio. No drift-outcome correlation is present.

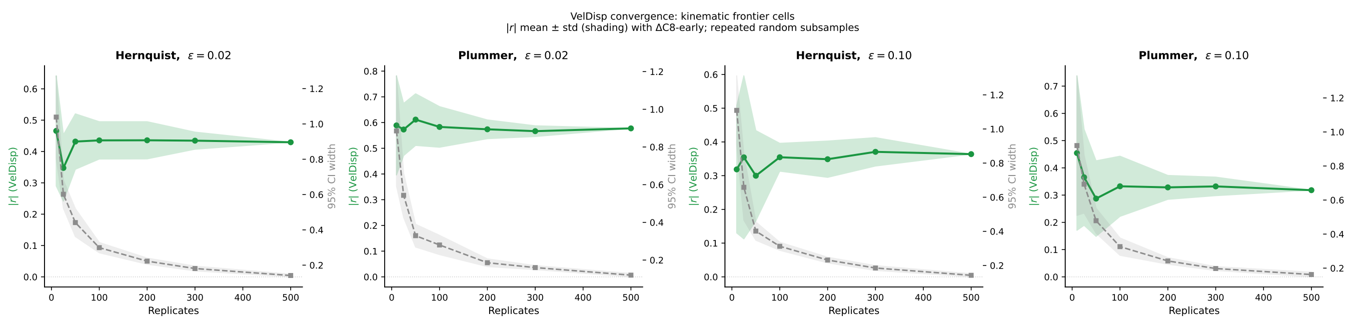

Figure 15. VelDisp convergence for kinematic frontier cells (Hernquist and Plummer at $\varepsilon = 0.02$ and 0.10). Green: $|r|$(VelDisp) mean $\pm$ std across repeated random subsamples as a function of replicate count. Grey dashed: 95% CI width (right axis). Both $|r|$ and CI width stabilise by ~ 200 replicates, confirming that the battery's 500 replicates are sufficient.

APPENDIX

A. VALIDATION AND DIAGNOSTIC FIGURES

A.1. Energy-drift outlier analysis

Of the 70,000 direct-isolated runs, 71 have a fractional energy drift $|\Delta E/E_0| \geq 0.1$ and 7 have drift ≥ 1.0. The single worst outlier (drift ≈ 17) is a Hernquist run at $N = 2048$, $\varepsilon = 0.03$; this corresponds to the rightmost point in the drift distribution panel of Figure 14. Outlier runs are concentrated in small-N Hernquist and Plummer realisations at low ε, where the softened cusp is marginally resolved and close two-body encounters occasionally inject energy.

To test quality-cut sensitivity, the full battery analysis was repeated after excluding runs with drift ≥ 0.1 (removing 71 runs, $< 0.1\%$ of the direct-isolated sample). No categorical verdict (COARSE/FINE_KIN/TIE) changed under this cut; all winner-gap confidence intervals shifted by less than their own half-width. A stricter cut at drift ≥ 0.01 (preserving 99.9% of runs) likewise leaves all verdicts unchanged. The conclusions of this paper are therefore insensitive to the small fraction of high-drift runs.

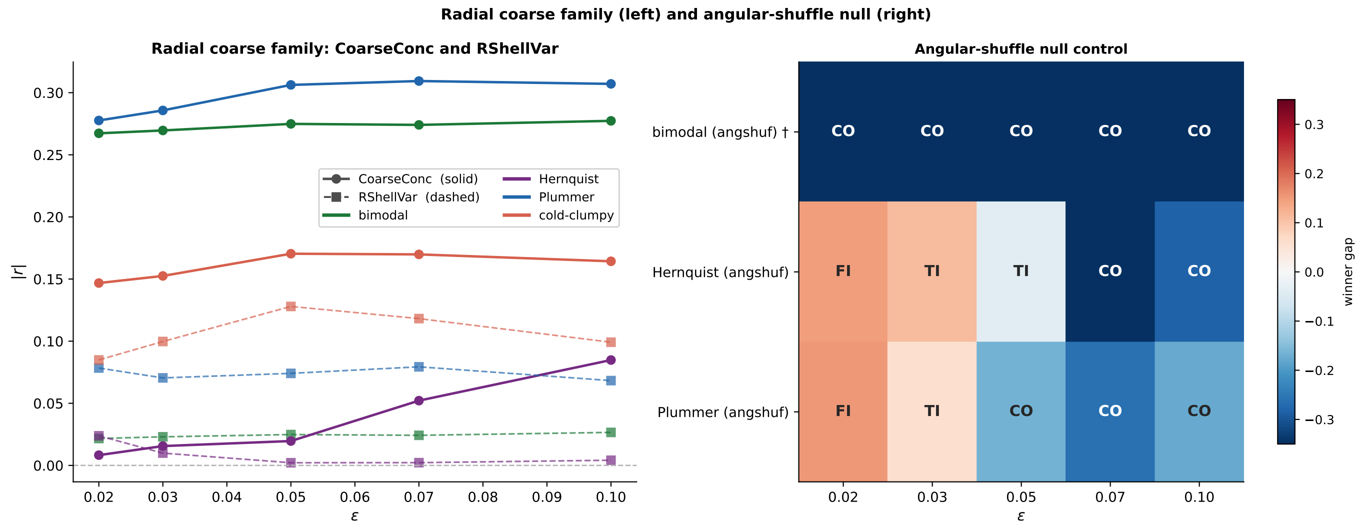

Figure 16. Diagnostic controls at $N = 1024$, direct-isolated. *Left*: $|r|$ for two radial coarse observables—CoarseConc (solid, circles) and CoarseRShellVar (dashed, squares)—by IC family and ε. Both remain weaker than the grid-based coarse predictor, confirming that the coarse advantage is not an artefact of grid geometry. *Right*: angular-shuffle null control. Colour encodes winner-gap mean; cell labels show the verdict (CO/FI/TI). Bimodal angular-shuffle (†) preserves inter-clump separation, so COARSE there is expected. For Hernquist and Plummer, the fine signal collapses under shuffling at large ε, confirming structure-dependence.

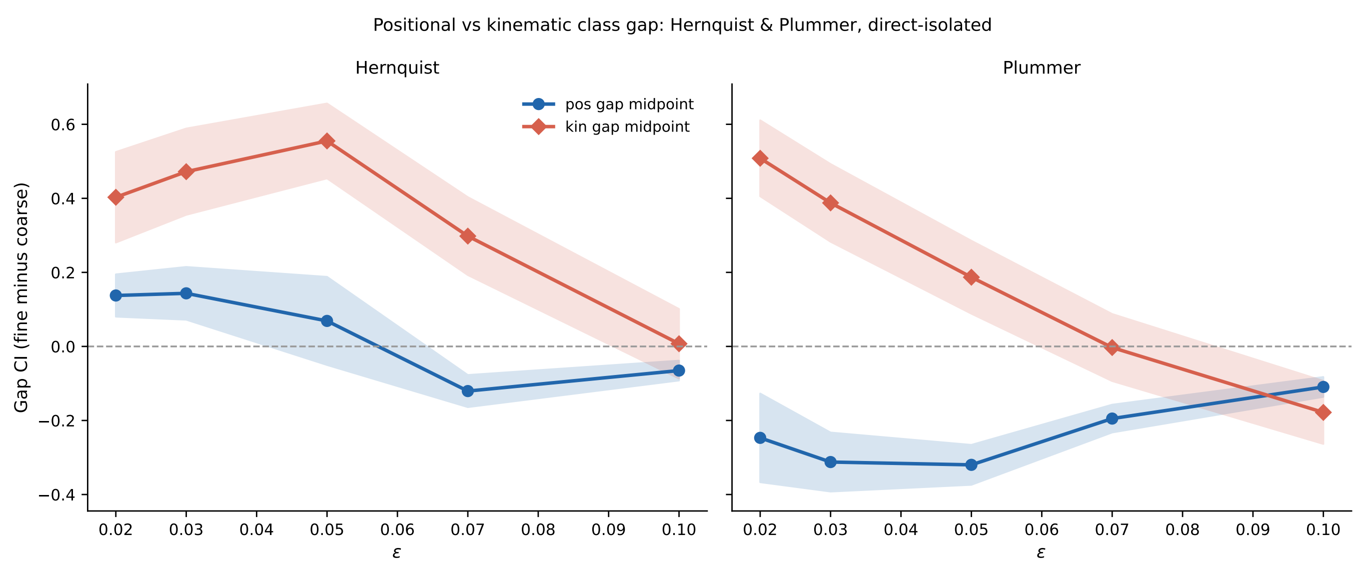

Figure 17. Class-level gap summary: distribution of $|r_{\rm fine}| - |r_{\rm coarse}|$ grouped by observable class (positional fine, kinematic fine) across all IC families and ε values at $N = 1024$, direct-isolated. Kinematic fine (VelDisp) is the only class whose gap distribution extends above zero for concentrated profiles at low ε. Positional fine classes remain below or near zero in all tested cells.

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